

Part Two

Thermal Physics

Thermal motion (热运动) is the random motions of a large number (10^{23}) of molecules (particles).

Thermal properties macro global

What is heat?

Temperature $\sim$ average kinetic energy?

Are all particles at rest at zero temperature?

Are perpetual motion machines possible?

...

general laws:

**conservation of energy,
law about irreversible processes,
phase change ...**

Fundamental level:

Arrow of time

black night sky paradox

microwave background radiation

relativistic effect

black hole ...

Thermodynamics

based on a small number of laws (zeroth, first, second and third laws) and mathematics

PVT...

generality and reliability

× details

分子动理学(理论)

Kinetic theory of gases and Statistical mechanics

assumption about the structure of matter and mathematical approximation

insight into details

fluctuation, critical phenomena

Chapter 10

Temperature

10.1 Equilibrium state

System: **finite, macroscopic** object
regardless of its quantity of matter,
region of space it occupied...

gases, rubber band, metal, magnetic material
(ferromagnet, ferrite), ferroelectrics,
liquid crystal, surface and film,
chemical substance ...

System _____ (physical or imaginary) **boundary**
_____ **surrounding** (medium, environment)

reservoir or heat bath

system	exchange
isolated	×
closed	energy
open	energy & matter

Phase is a homogeneous (chemical composition
and physical state) part of system
coexistence, phase transition

To describe equilibrium state, we introduce **state variables**. They could be

- mechanical, p ,
- electromagnetic, E, P, B, M ,
- geometric, V, L ,
- chemical, n, x_i
-

F is **extensive** if $F (n \text{ systems}) = nF (1 \text{ system})$

广延量 V, n

F is **intensive** if $F (n \text{ systems}) = F (1 \text{ system})$

强度量 p

Equilibrium state

Macroscopic variables have definite values that remain constant so long as the external conditions are unchanged.

It is a dynamic equilibrium (fluctuations are denied).

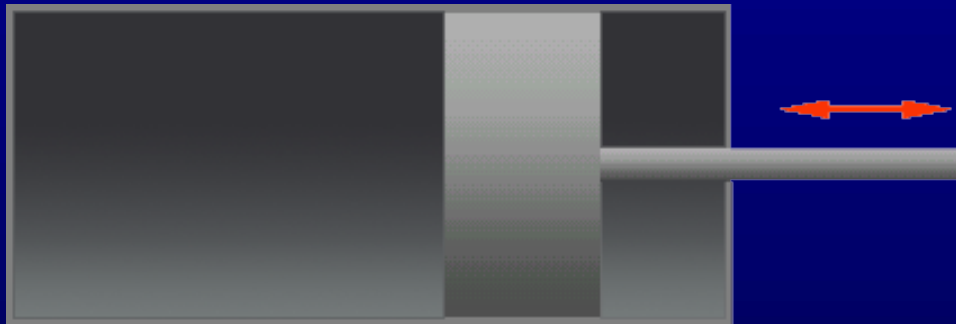
No flux is involved. (otherwise 'stationary' state)

System may be of piecewise uniformity

relaxation time

弛豫时间

- surrounding changed,
- system deviates from equilibrium
- after a time τ , system reaches a new equilibrium state



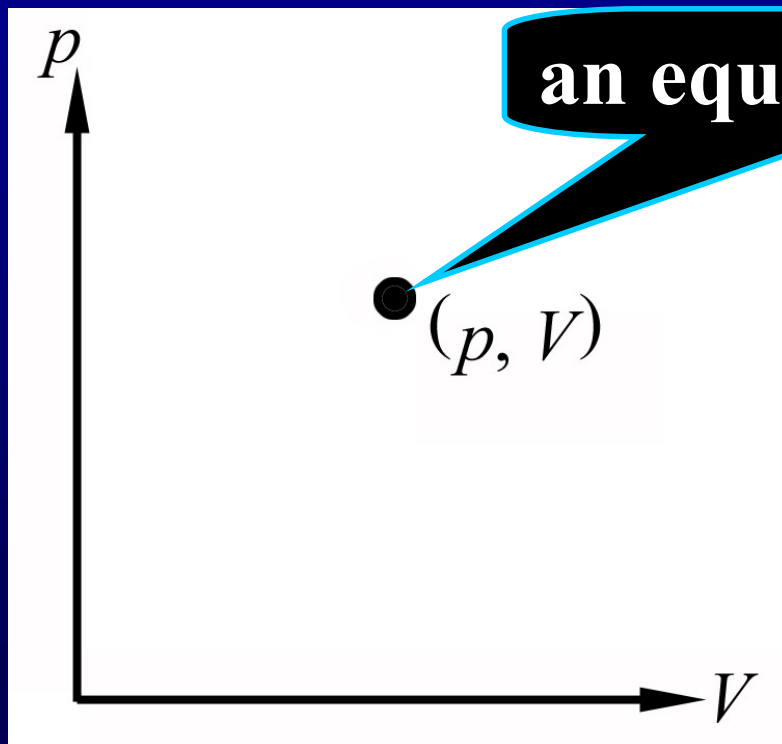
$$L(0.1\text{m} \sim 1\text{m}) \quad v_{\text{sound}} \sim 300\text{ m/s}$$

$$\tau \propto \frac{L}{v} \propto \frac{0.3\text{m}}{300\text{ m/s}} = 10^{-3}\text{ s}$$

functions of state

$$pV = nRT$$

choose (p, V) as variables, and T as function

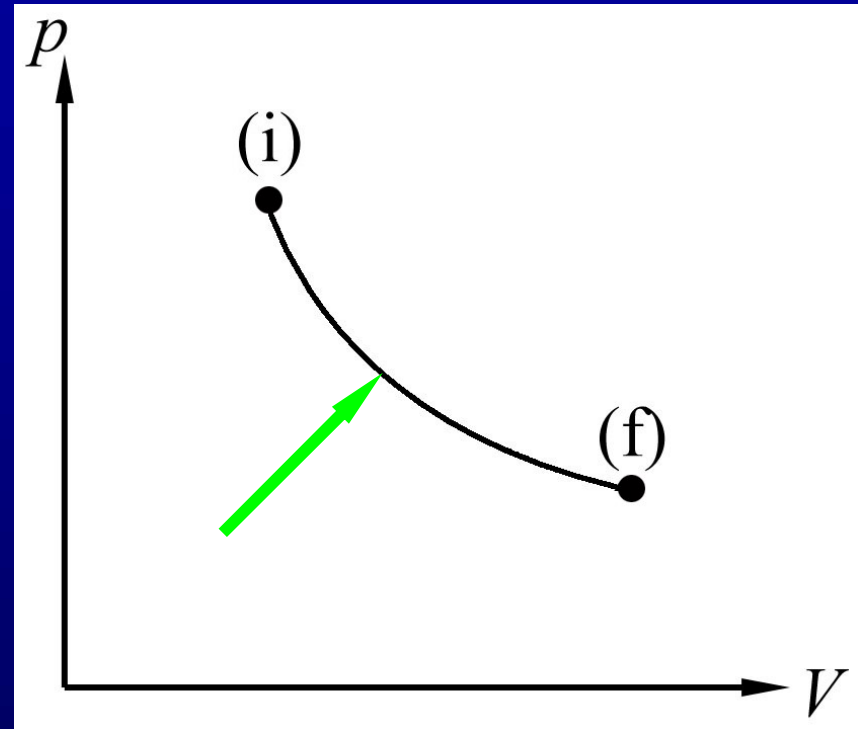
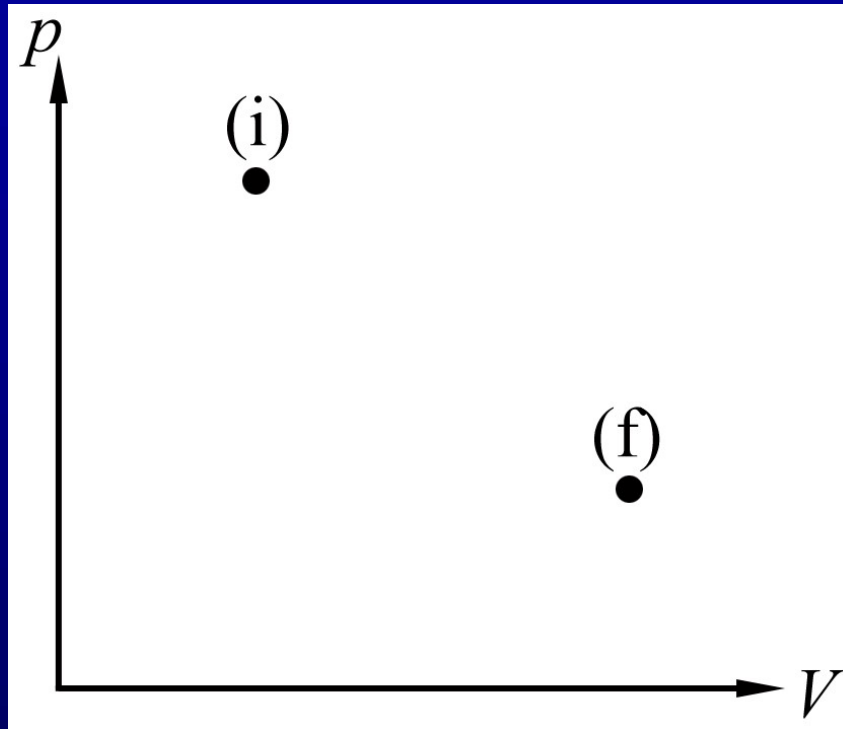


an equilibrium state

State diagram

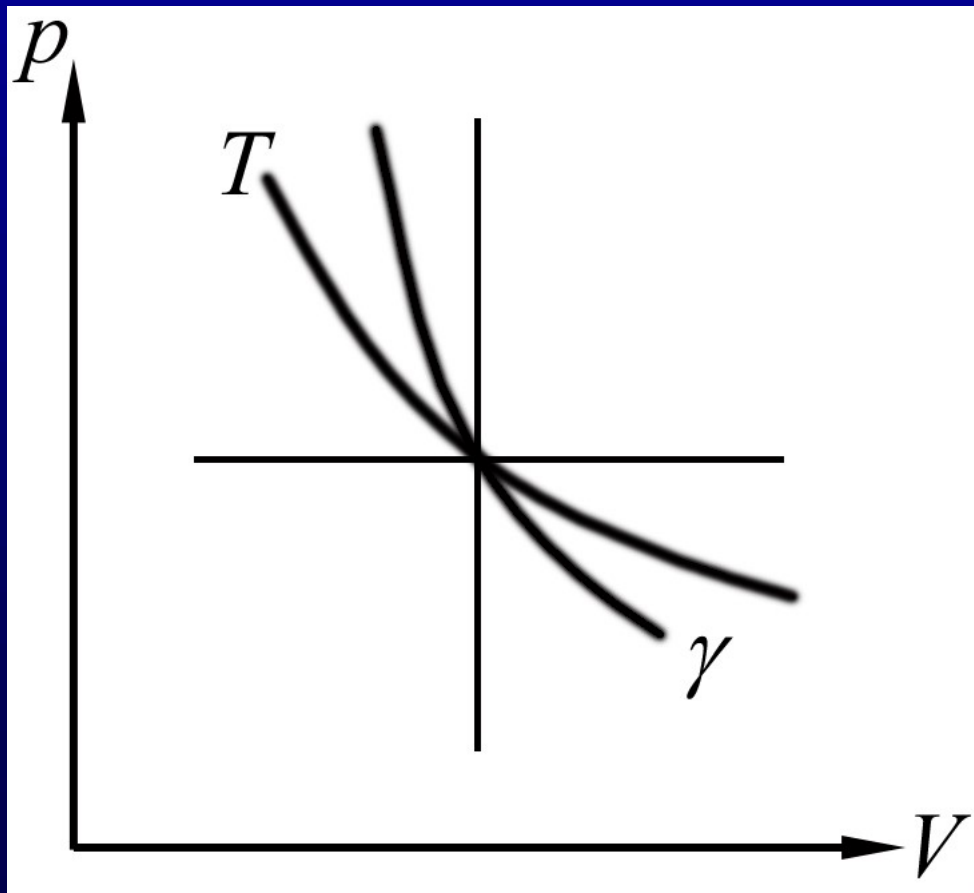
p, V thermodynamic
coordinates

Change of state & history of the change



$$t_{\text{piston}} = 0.1\text{s}, \quad \tau = 10^{-3}\text{s}, \quad t \gg \tau$$

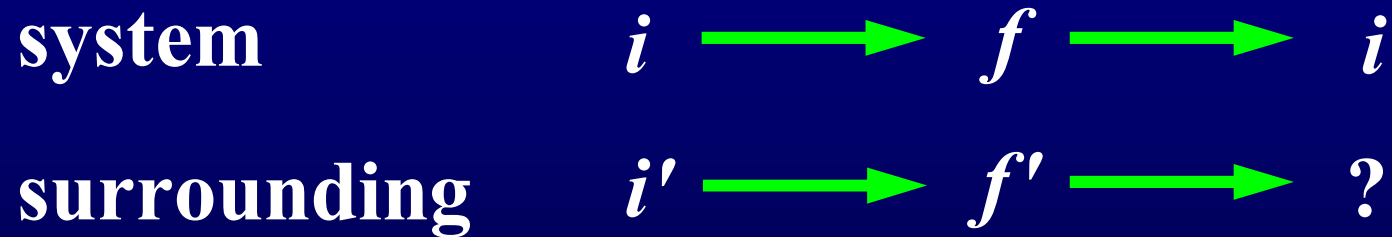
Quasi-static process is performed slowly enough



Processes:

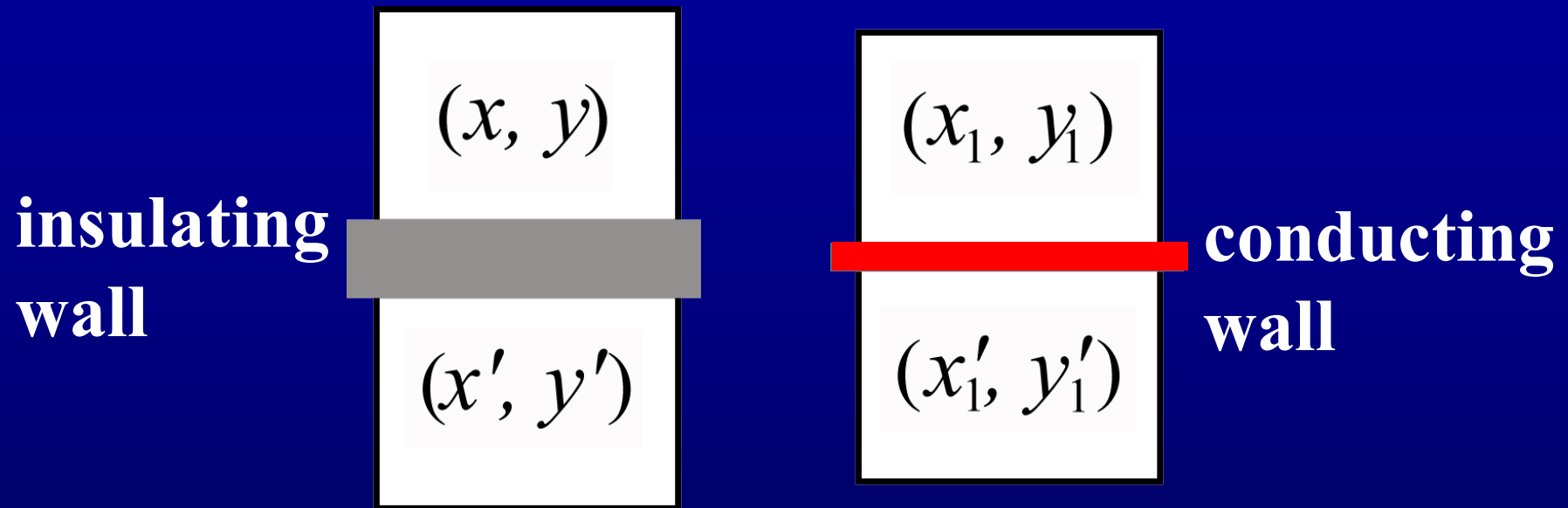
- constant pressure (isobaric)
- constant volume (isochoric)
- isothermal
- isentropic (adiabatic)

While no-dissipative, quasi-static process is reversible,
all natural processes (spontaneous processes) are irreversible.



If i' , system experiences a reversible process

10.2 Thermal equilibrium and temperature



reach an equilibrium state
of the **combined system**
thermal contact

The zeroth law of thermodynamics reads:

- *Two systems, each in **thermal equilibrium** with a third (thermometer) are in thermal equilibrium with each other.*

Zeroth law: C.Carathéodory (1909),
(kar-a-thee-a-DOR-ee)
(enters textbook) R H Fowler (1939)

The **temperature** of a system is a property that determines whether or not the system is in **thermal equilibrium** with other system.

macroscopic concept. macroscopic quantity

no restriction on the **range**, built-in sense of **direction**, and even **positive or negative**.

10.3 Empirical temperature scales

- the thermometric property of substance

Mechanical	p of a gas kept at constant V tension J of a rubber band
geometric	V of a gas kept at constant p V of liquid e.g. mercury
Electrical	resistance R of a wire EMF of a thermocouple

$$T(X) = aX$$

- a standard fixed point

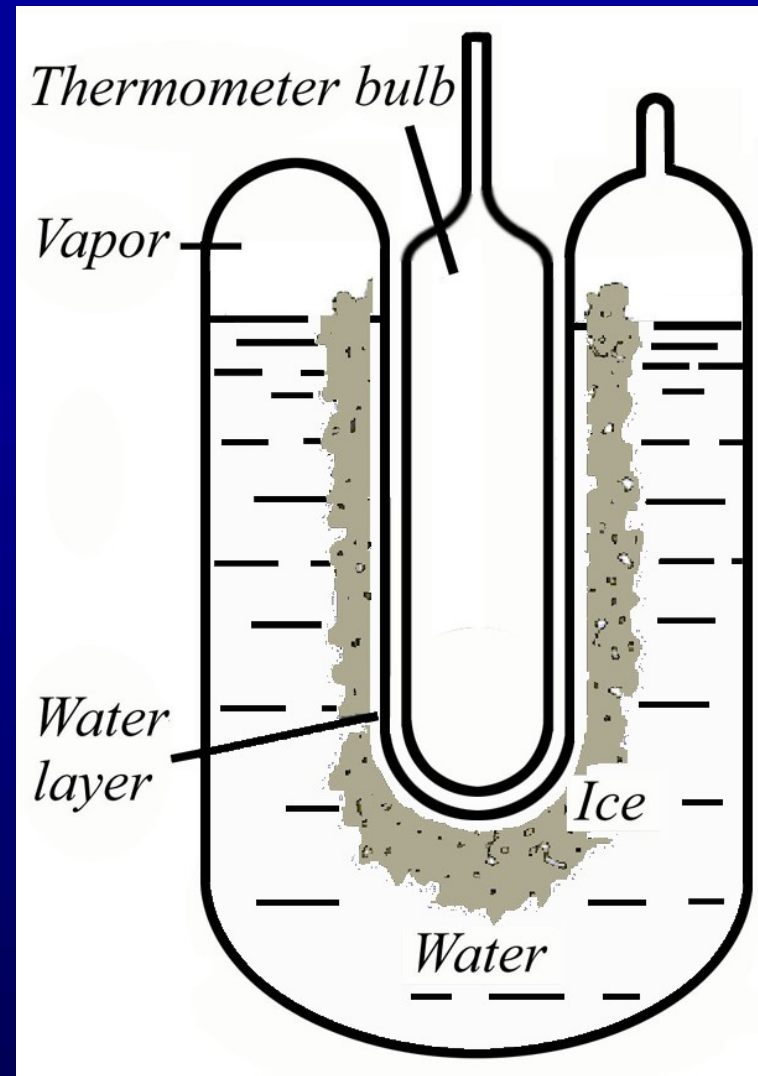
$$T(X) = aX$$

$$T_{\text{tr}} = 273.16\text{K} \quad a = \frac{T_{\text{tr}}}{X_{\text{tr}}}$$

$$\frac{T(X)}{273.16\text{K}} = \frac{X}{X_{\text{tr}}}$$

$$T(R) = 273.16\text{K} \frac{R}{R_{\text{tr}}}$$

$$T(V) = 273.16\text{K} \frac{V}{V_{\text{tr}}} (\text{constant pressure})$$



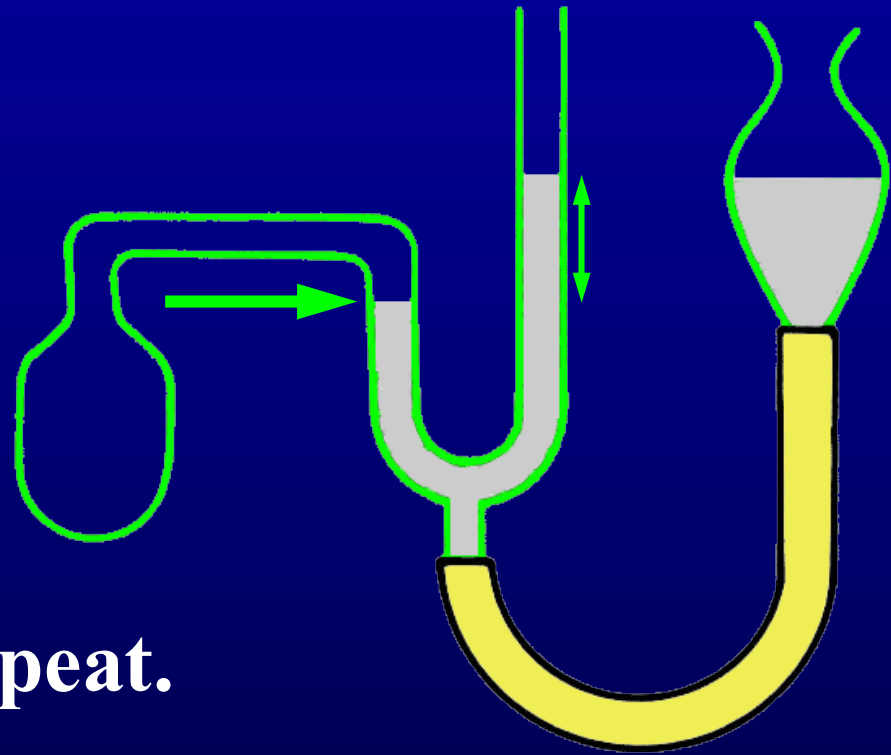
$$T = 273.16\text{K} \frac{p}{p_{\text{tr}}} (\text{constant volume})$$

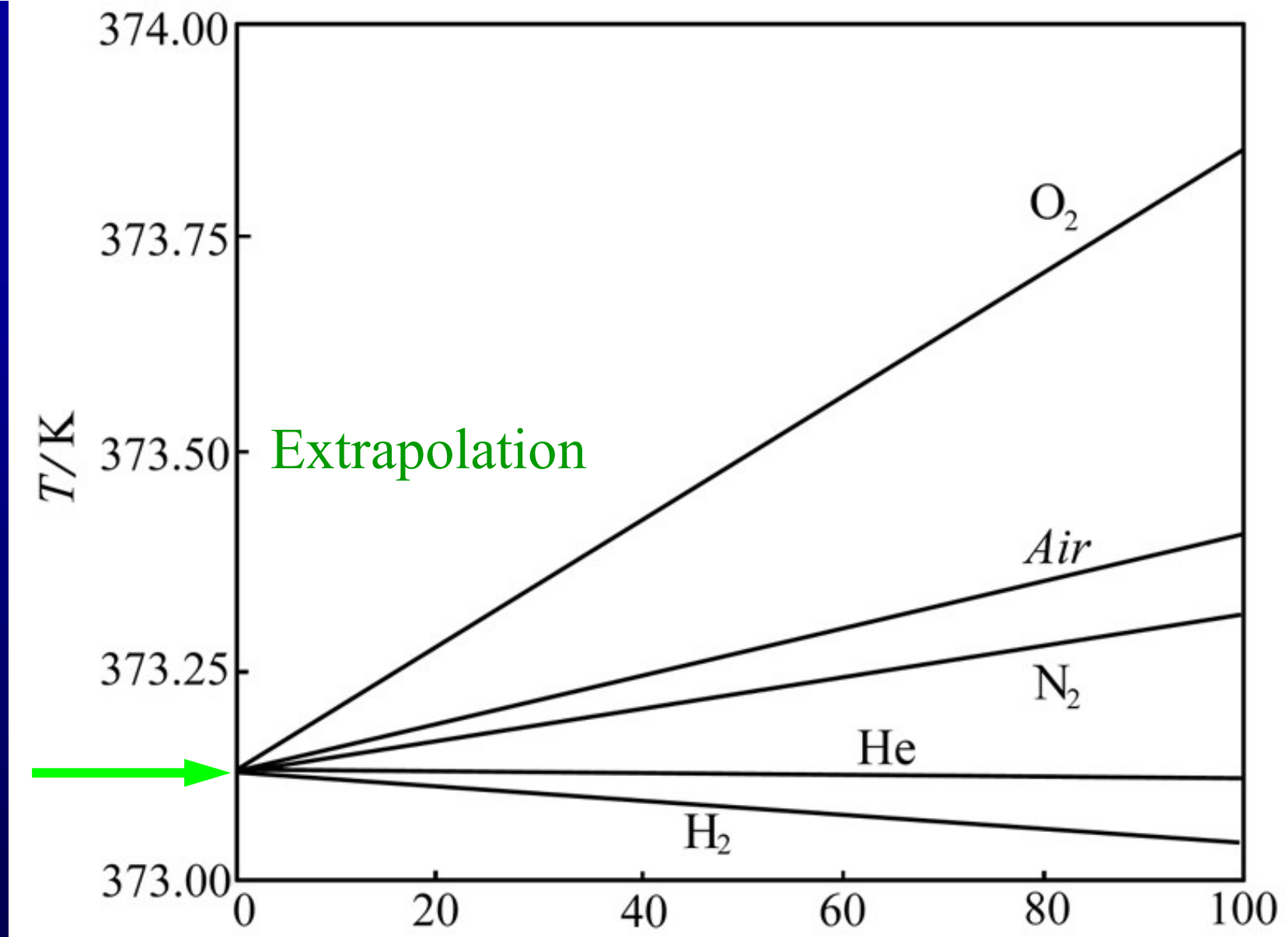
$$p_{\text{tr}} \quad p_{\text{steam}} \longrightarrow T_{\text{steam}}$$

**Reduce the amount
of gas, repeat**

$$(T_{\text{steam}}, p_{\text{tr}})$$

For different gases, repeat.





$p_{tr} / 1333.32 \text{ Pa}$

Ideal gas temperature scale:

$$T = 273.16\text{K} \lim_{p_{\text{tr}} \rightarrow 0} \left(\frac{p}{p_{\text{tr}}} \right)$$

Celsius temperature:(before 1948, T_c)

$$T_C = T_{\text{ideal gas}} - 273.15$$

Fahrenheit temperature

$$T_F = 32^\circ + \frac{9}{5} T_C$$

Practical Scale IPTS-68(75)

thermodynamic scale or Kelvin scale

**independent of the thermometric properties
of any particular substance**

$$5 \times 10^{-8} \text{ K} \sim 5 \times 10^8 \text{ K}$$

$$30 \text{ PK} = 30 \times 10^{-12} \text{ K}$$

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10.4 Equation of state

- based on experiments.
- as accurate as experiments
- valid within the range of values measured.
- In quasi-static process, equation of state exists.
- may be deduced from kinetic theory or statistical physics upon some assumption

For solid and liquid

coefficient of linear expansion is defined as

$$\alpha_\ell = \frac{1}{L} \left(\frac{\partial L}{\partial T} \right)_p$$

T, p variables

For isotropic material

$$\alpha_V = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p = \frac{1}{L^3} \left(\frac{\partial L^3}{\partial T} \right)_p = 3\alpha_\ell$$

$$\sim 10^{-7}/\text{K} \sim 10^{-5}/\text{K}$$

$$\frac{\Delta V}{V} \approx \alpha_V \Delta T \approx 10^{-5} \sim 10^{-3}$$

Isothermal compressibility is defined as

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

$$\sim 10^{-10} / \text{Pa}$$

$$\left| \frac{\Delta V}{V} \right| \approx \kappa_T \Delta p \approx 10^{-5}$$

$$1 \text{ atm} = 101325 \text{ Pa}$$

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

$$1 \text{ bar} = 10^5 \text{ Pa}$$

$$\frac{dV}{V} = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p dT + \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T dp$$

$$= \alpha_V(T) dT - \kappa_T(p) dp$$

$$\approx \alpha_V dT - \kappa_T dp$$

$$\ln \frac{V}{V_0} = \alpha_V \Delta T - \kappa_T \Delta p$$

$$V = V_0 \exp(\alpha_V \Delta T - \kappa_T \Delta p)$$

$$V \approx V_0 (1 + \alpha_V \Delta T - \kappa_T \Delta p)$$

Equation of state of ideal gas

$$pV = nRT \quad R = 8.31 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

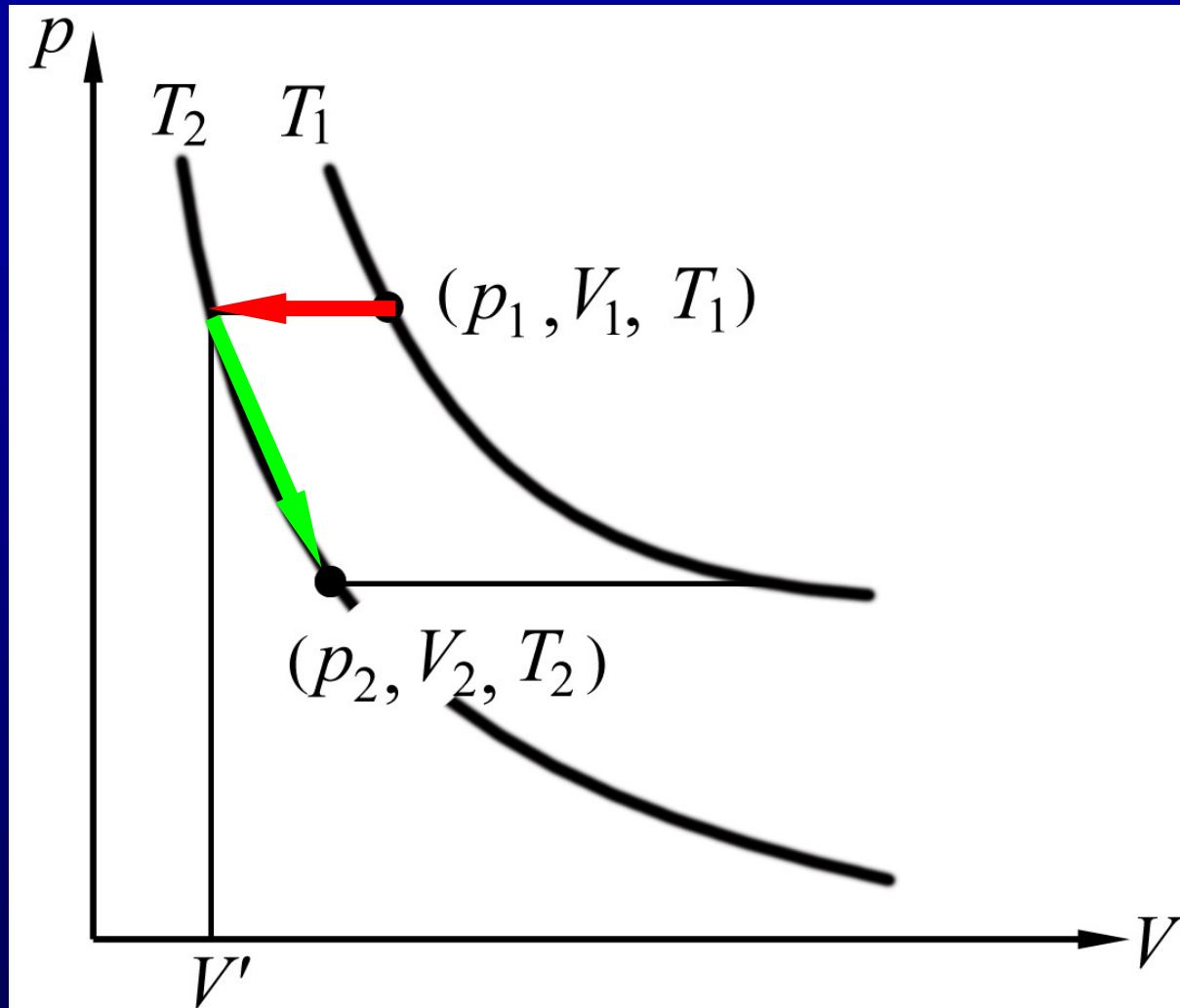
can be derived from

Boyle-Mariotte law

$$pV = C(T)$$

and gas constant pressure (empirical) **scale**

$$\frac{V}{T} = D(p) \quad T(V) = 273.16 \text{ K} \frac{V}{V_{\text{tr}}}$$



$$\frac{T_1}{V_1} = \frac{T_2}{V'}$$

$$p_1 V' = p_2 V_2$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Avogadro's law reads:

**under same temperature and pressure, gases of equal volume contain same number of molecules,
or**

under same temperature and pressure, molar volume of gas is same.

$$T_0 = 273.15\text{K}, \quad p_0 = 101325\text{Pa}$$

$$V_{m0} = 22.414 \times 10^{-3} \text{ m}^3/\text{mol}$$

 **Gas constant R**

Boyle-Mariotte law and Avogadro's law apply to real gas approximately.

dilute $\longrightarrow$ $\sim$ ideal gas

**microscopic viewpoint: non-interacting system.
range of interaction $\ll$ average distance between
its molecules.**

without interaction, it can never be liquified.

For Real gases, Virial expansion applies

$$pV = nRT \left[1 + B(T)p + C(T)p^2 + \dots \right]$$

$$pV = nRT \left[1 + B'(T)\frac{n}{V} + C'(T)\left(\frac{n}{V}\right)^2 + \dots \right]$$

B, C (B', C') ~ second, third Virial coefficient

Van der Waals' equation (1873)

$$\left[p + a\left(\frac{n}{V}\right)^2 \right] (V - nb) = nRT$$

Assignment: 10.1, 10.4, 10.5